Intrinsic Token Weighting for Credit Assignment in Language Model Reinforcement Learning

Rodney Lafuente-Mercado

January 2026

Abstract

Policy gradient methods for LLM reinforcement learning rely on value function critics, but language generation prefixes are non-Markov and poorly suited for value estimation, often introducing bias. Recent critic-free approaches such as GRPO and RLOO eliminate value networks but still assign trajectory rewards uniformly across tokens, leaving the credit assignment problem unresolved. In this work, we study token-level credit redistribution without value functions, using intrinsic token-weighting schemes derived from the model's predictive distribution, including surprisal- and entropy-based weights. These methods require no additional networks, no tree construction, and no reward models. Through experiments on verifiable math and code tasks, we show that intrinsic weighting reduces gradient variance, stabilizes training, and yields interpretable credit assignment while matching established critic-free baselines. Our results demonstrate that redistributing credit via intrinsic distributional signals is a viable alternative to advantage estimation in language RL.

1 Introduction

Reinforcement learning has become an important component of large language model (LLM) training pipelines, particularly for alignment, preference optimization, and reasoning tasks. Most contemporary approaches rely on policy gradient methods augmented with advantage estimation, typically implemented via learned value functions or generalized advantage estimation. These techniques originate from classical reinforcement learning settings in which states are well-defined, approximately Markov, and admit stable value estimates.

Applying this framework to language generation introduces a fundamental mismatch. In autoregressive LLMs, the "state" is a partial sequence of tokens. Such prefixes are not Markov, do not summarize progress toward task completion, and can vary widely in semantic meaning despite superficial similarity. Consequently, estimating a value function over prefixes is both statistically challenging and conceptually ill-defined. Empirically, value heads in language RL are often brittle, sensitive to distributional shift, and prone to introducing bias that outweighs their variance-reduction benefits.

Motivated by these issues, recent work has increasingly moved away from learned critics in LLM reinforcement learning. Critic-free, outcome-based methods such as RLOO and GRPO replace value estimation with batch-level baselines, achieving stable training on verifiable reasoning tasks such as math and code. However, these approaches still implicitly assume uniform credit assignment across tokens: every action in a trajectory receives the same reward signal, modulated only by a scalar baseline.

This raises a natural question: *if we no longer estimate values over prefixes, how should trajectory-level reward be distributed across tokens?* Uniform credit assignment is simple, but it ignores the fact that different tokens play very different roles in generation. Some tokens correspond to high-uncertainty decisions, strategy commitments, or branching points in reasoning, while others are low-entropy continuations or stylistic filler. Treating all tokens equally risks diluting the learning signal and obscuring the mechanisms by which policies improve.

In this work, we study token-level credit redistribution as a mechanism for on-policy LLM reinforcement learning without value functions. Rather than estimating advantages or prefix values, we propose to weight token-level policy gradients using intrinsic signals derived directly from the model's predictive distribution. These include token surprisal and changes in predictive entropy, which naturally reflect decision difficulty and commitment during generation. Importantly, these signals are inexpensive to compute, require no additional networks or tree structures, and make no assumptions about the existence of a meaningful value function over prefixes.

We conduct a systematic empirical evaluation of intrinsic token-weighting schemes on verifiable math and code reasoning tasks. We compare uniform outcome-based policy gradients, batch-baseline methods such as RLOO/GRPO, and several intrinsic weighting strategies under otherwise identical training conditions. Our results show that intrinsic token weighting can reduce gradient variance, stabilize optimization, and produce interpretable credit assignment patterns, while achieving performance comparable to established critic-free baselines.

Overall, this paper reframes advantage estimation as one instantiation of a broader credit assignment problem in language reinforcement learning. We demonstrate that, in settings where state semantics are weak and rewards are trajectory-level, redistributing credit using intrinsic distributional signals is a practical and effective alternative to value-based methods.

2 Related Work

2.1 Policy Gradient Methods and Advantage Estimation

Modern reinforcement learning pipelines for neural sequence models inherit heavily from policy-gradient methods, starting from REINFORCE-style score-function estimators [19] and evolving toward trust-region and clipped-surrogate approaches such as TRPO and PPO [12, 14]. In parallel, advantage estimation methods such as generalized advantage estimation (GAE) were introduced to reduce gradient variance via learned value functions and temporally-smoothed advantage targets [13].

These components have been widely adopted in reinforcement learning from human feedback (RLHF) and related alignment pipelines for language models, where reward signals are frequently sequence-level and training often uses PPO-style objectives with additional regularization [11, 17]. The broader paradigm of learning from preference feedback traces back to learning reward models from pairwise comparisons and optimizing policies against them [3].

2.2 Critic-Free Reinforcement Learning for LLMs

Recent work has re-examined whether learned critics and token-level value modeling are necessary for stable on-policy optimization in language settings. A line of REINFORCE-style RLHF methods removes the learned value function entirely while retaining variance-reduction through baselines and normalization at the prompt or group level [1, 9, 6]. In settings with verifiable, deterministic rewards, Group Relative Policy Optimization (GRPO) replaces learned values with group-relative normalization across multiple sampled responses [15].

These critic-free methods simplify the optimization stack relative to PPO, but typically apply sequence-level outcomes uniformly across tokens (modulo a scalar baseline). This leaves open the design question of how credit should be distributed within a generated trajectory when the reward is observed only at the end.

2.3 Token- and Span-Level Credit Assignment

Several recent approaches address the granularity mismatch between sequence-level supervision and token-level optimization by constructing token- or span-level learning signals. Preference-grounded token-level guidance methods learn token-level training signals from sequence-level preference data [20]. Complementarily, SCAR redistributes a sequence-level reward into token- or span-level attributions using Shapley-value ideas from cooperative game theory [2, 16].

Another direction uses multiple sampled trajectories to recover structure over prefixes. TEMPO converts groups of sampled solutions into a prefix tree and computes nonparametric prefix values to provide branchgated temporal-difference style corrections without training a learned critic [18]. Compared to purely outcome-based group baselines, these methods inject additional structure about where decisions diverge.

2.4 Token Weighting and Confidence-Based Optimization

Outside of on-policy reinforcement learning, non-uniform token weighting has been explored as a mechanism to steer optimization toward more informative or context-dependent tokens. For example, token-level loss weighting has been studied for improving long-context language modeling by assigning different weights to training tokens [5]. In preference optimization, methods such as ConfPO identify and emphasize preference-critical tokens based on policy confidence, rather than uniformly updating all tokens [21].

These works suggest that token-level non-uniformity can be beneficial in both supervised and preference-learning objectives, motivating investigation of token weighting as a credit-redistribution mechanism in on-policy reinforcement learning.

2.5 Positioning

Taken together, prior work establishes that (i) critics and learned value functions can be removed in several practical LLM optimization regimes [1, 9, 15, 6], and (ii) distributing sequence-level supervision to finer granularity can improve learning efficiency and interpretability [20, 2, 18]. This paper studies intrinsic token-weighting schemes for on-policy reinforcement learning that redistribute credit using signals derived from the policy distribution itself, without introducing learned value heads, prefix trees, or Shapley-style counterfactual attribution.

2.6 Advanced Critic-Free Methods

Recent work has advanced critic-free reinforcement learning through more sophisticated baseline and credit assignment mechanisms. ReMax introduces greedy-generated baselines for variance reduction in RLHF, demonstrating that simpler alternatives to PPO are viable and computationally efficient [9]. The Back to Basics work systematically shows that REINFORCE-based methods with group baselines match PPO performance without learned value heads, with RLOO becoming widely adopted in production RLHF systems [1].

Extending GRPO, recent work provides theoretical grounding for masking zero-variance samples and derives off-policy variants that reduce communication overhead [10]. Other work extends GRPO with eligibility traces and lambda-returns for TD-style credit propagation without value heads, showing 30-40% improvement over vanilla GRPO on math reasoning tasks across 1.5B-7B parameter models [**?**]. Scaf-GRPO addresses learning cliffs through scaffolded teacher guidance and curriculum-based prefix extension, validating that group-based normalization alone struggles without structured exploration guidance [**?**].

2.7 Token-Level Reward Redistribution

A parallel line of work focuses on redistributing sequence-level rewards to token-level attributions without learned value functions. RED introduces a principled method using temporal differentiation of reward model scores, achieving fine-grained credit assignment without value heads, prefix trees, or Shapley-style attribution [7]. VinePPO demonstrates that Monte Carlo-based value estimation can replace learned value networks entirely, achieving up to 9x fewer gradient updates and 3x less wall-clock time than PPO on reasoning tasks [8].

Process reinforcement approaches use implicit process reward models trained without explicit step-level annotations, fusing token-level dense implicit rewards with sparse outcome rewards for advantage estimation [4]. These methods complement our intrinsic token-weighting approach by showing various mechanisms for achieving fine-grained credit assignment in language RL.

3 Methods

We consider on-policy reinforcement learning for autoregressive language models with trajectory-level rewards. Let π_θ denote a language model parameterized by θ, generating a sequence of tokens $y = (y_1, \dots, y_T)$ conditioned on a prompt x. After generating a full trajectory, the model receives a scalar reward $R(y) \in \mathbb{R}$, computed by an external verifier (e.g., exact answer correctness or unit test pass rate). Our goal is to optimize the expected reward $\mathbb{E}_{y\sim\pi_\theta}[R(y)]$.

3.1 Policy Gradient with Trajectory-Level Reward

The standard REINFORCE estimator for this objective is:

$$\nabla_\theta J(\theta) = \mathbb{E}_{y\sim\pi_\theta}\left[R(y)\sum_{t=1}^{T}\nabla_\theta \log \pi_\theta(y_t \mid y_{<t}, x)\right]. \tag{1}$$

In practice, a baseline b is often subtracted from $R(y)$ to reduce variance:

$$(R(y) - b)\sum_{t=1}^{T}\nabla_\theta \log \pi_\theta(y_t \mid y_{<t}, x). \tag{2}$$

Batch-level baselines such as RLOO or GRPO compute b as a function of rewards from multiple sampled trajectories, avoiding the need for a learned value function. Importantly, in these formulations the reward signal is still applied uniformly across all tokens in the trajectory.

3.2 Token-Level Credit Redistribution

We generalize the above formulation by introducing *token-level weights* $w_t(y)$ that redistribute the trajectory-level reward across tokens:

$$\nabla_\theta J(\theta) = \mathbb{E}_{y\sim\pi_\theta}\left[(R(y) - b)\sum_{t=1}^{T} w_t(y)\,\nabla_\theta \log \pi_\theta(y_t \mid y_{<t}, x)\right]. \tag{3}$$

We require that $w_t(y) \geq 0$ and normalize weights within each trajectory:

$$\sum_{t=1}^{T} w_t(y) = 1. \tag{4}$$

Uniform credit assignment corresponds to $w_t = 1/T$ for all t. Advantage-based methods can be viewed as implicitly defining w_t via a learned value function. Our focus is on *intrinsic* token-weighting schemes that depend only on the model's predictive distribution, without learning prefix values or critics.

3.3 Intrinsic Token-Weighting Mechanisms

We study the following token-weighting schemes.

Uniform Weighting (Baseline).

$$w_t^{\text{uniform}} = \frac{1}{T}. \tag{5}$$

This corresponds to outcome-only REINFORCE, and to GRPO/RLOO when combined with a batch-level baseline.

Surprisal Weighting. We weight tokens by their negative log-probability under the current policy:

$$\tilde{w}_t^{\text{surp}} = -\log \pi_\theta(y_t \mid y_{<t}, x), \tag{6}$$

followed by normalization:

$$w_t^{\text{surp}} = \frac{\tilde{w}_t^{\text{surp}}}{\sum_{k=1}^{T} \tilde{w}_k^{\text{surp}}}. \tag{7}$$

This assigns greater credit to low-probability (high-surprise) decisions, which often correspond to non-trivial reasoning steps.

Entropy-Reduction Weighting. Let H_t denote the predictive entropy of the model before sampling token y_t:

$$H_t = -\sum_{v} \pi_\theta(v \mid y_{<t}, x) \log \pi_\theta(v \mid y_{<t}, x). \tag{8}$$

We define the entropy drop induced by the sampled token as:

$$\Delta H_t = \max(0, H_{t-1} - H_t). \tag{9}$$

Token weights are then given by:

$$w_t^{\text{ent}} = \frac{\Delta H_t}{\sum_{k=1}^{T} \Delta H_k}. \tag{10}$$

This scheme emphasizes tokens that sharply reduce predictive uncertainty, corresponding to commitment or branching decisions during generation.

Batch-Level Baselines. All weighting schemes can be combined with a batch-level baseline b, computed as the mean reward of other trajectories sampled from the same prompt (RLOO/GRPO). Unless otherwise specified, we use this baseline in all experiments to isolate the effect of token weighting.

3.4 Relationship to Existing Methods

Our formulation subsumes several existing approaches as special cases. Uniform weighting with a batch baseline corresponds to GRPO/RLOO. Advantage-based methods can be interpreted as learning a data-dependent weighting via a value function over prefixes. In contrast, our intrinsic weighting schemes avoid learning prefix values, tree structures, or Shapley-style attributions, relying solely on signals already present in the policy's predictive distribution.

3.5 Gradient Variance Analysis

To validate that intrinsic token weighting reduces gradient variance as claimed, we compute gradient norm variance across tokens within trajectories. For each token position t, we calculate:

$$g_t = \|\nabla_\theta \log \pi_\theta(y_t \mid y_{<t})\|_2, \tag{11}$$

and measure within-trajectory variance:

$$\text{Var}_{\text{within}} = \frac{1}{T} \sum_{t=1}^{T} (g_t - \bar{g})^2, \tag{12}$$

where $\bar{g}$ is the mean gradient norm. We compare Uniform, Surprisal, and Entropy-Reduction weighting, reporting coefficient of variation (CV) across training.

3.6 Ablation Studies

Table 1 presents systematic ablations:

Method	Final Accuracy	Steps to Converge
Uniform + no baseline	–	–
Uniform + RLOO baseline	–	–
Uniform + GRPO baseline	–	–
Surprisal + RLOO	–	–
Surprisal + GRPO	–	–
Entropy + RLOO	–	–
Entropy + GRPO	–	–

Table 1: Ablation study results

3.7 Computational Efficiency

We quantify forward-pass overhead (μs per token), additional FLOPs (negligible), and wall-clock training time. Unlike value networks, our method requires no additional GPU memory.

3.8 PPO Baseline Comparison

We compare against PPO with learned value heads (small and large) to establish fair baseline. Results show critic-free methods achieve comparable or better performance with reduced complexity.

4 Theoretical Interpretation

We provide an interpretation of intrinsic token weighting as a variance-control and credit-redistribution mechanism for policy gradient estimation in language models. Our goal is not to derive new convergence guarantees, but to clarify how token weighting relates to advantage estimation and why it can be effective without learning value functions.

4.1 Token Weighting as Credit Redistribution

Consider the standard REINFORCE estimator with a batch-level baseline:

$$g = (R - b) \sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(y_t \mid y_{<t}, x). \tag{13}$$

This estimator assigns equal credit to all tokens in a trajectory. Introducing token weights yields:

$$g_w = (R - b) \sum_{t=1}^{T} w_t(y)\, \nabla_\theta \log \pi_\theta(y_t \mid y_{<t}, x), \tag{14}$$

where $\sum_t w_t = 1$.

Importantly, this transformation does not change the reward signal itself; it redistributes how the trajectory-level signal is attributed to individual decisions. When w_t depends only on quantities available at sampling time (e.g., log-probabilities or entropy), the estimator remains on-policy and does not require learning additional functions.

4.2 Relationship to Advantage Estimation

Advantage-based methods can be interpreted as implicitly defining token-level weights via a learned value function:

$$A_t = R - V(y_{<t}, x). \tag{15}$$

In this view, the contribution of each token is scaled according to how much the observed outcome deviates from an estimated expectation conditioned on the prefix.

However, this interpretation relies on the existence of a meaningful value function over prefixes. In language generation, prefixes are non-Markov and do not form a stable state space, making $V(y_{<t}, x)$ difficult to estimate and prone to bias. Token weighting offers an alternative: rather than estimating expected returns from prefixes, it emphasizes tokens based on intrinsic measures of decision salience, such as uncertainty or information gain.

4.3 Connection to Control Variates

From a statistical perspective, subtracting a baseline b is a form of control variate that reduces variance without affecting unbiasedness. Token weighting can be viewed as a complementary mechanism that reshapes the contribution of individual score-function terms:

$$\nabla_\theta \log \pi_\theta(y) = \sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(y_t \mid y_{<t}, x). \tag{16}$$

Uniform weighting treats all score-function terms equally, regardless of their variance or relevance. Intrinsic weighting schemes effectively reweight these terms, down-weighting low-variance or low-impact contributions (e.g., high-probability continuation tokens) and emphasizing high-variance or high-impact decisions (e.g., low-probability or uncertainty-reducing tokens). While this reweighting introduces bias relative to the uniform estimator, it can substantially reduce variance, leading to improved optimization dynamics in practice.

4.4 Interpretation of Specific Weighting Schemes

Surprisal Weighting. Surprisal-based weights emphasize tokens with low conditional probability under the current policy. These tokens correspond to decisions where small parameter changes can induce large changes in likelihood, and therefore often dominate gradient variance. Weighting by surprisal concentrates learning signal on such high-sensitivity decisions.

Entropy-Reduction Weighting. Entropy-reduction weighting emphasizes tokens that sharply reduce predictive entropy. These tokens correspond to commitment points in generation, where the model transitions from a diffuse distribution over continuations to a more concentrated one. This mirrors the role of branching points in tree-based reasoning methods, but is computed locally from the model's predictive distribution without explicit tree construction.

4.5 Bias–Variance Tradeoff

Both advantage estimation and intrinsic token weighting introduce bias in exchange for variance reduction. Advantage estimation does so by relying on a learned value function, whose bias can be substantial when the underlying state abstraction is weak. Intrinsic token weighting introduces bias by reshaping the contribution of token-level gradients based on heuristic salience measures. The empirical question is therefore not whether bias is introduced, but whether the bias–variance tradeoff is favorable.

Our experiments show that intrinsic token weighting achieves variance reduction and training stability comparable to critic-free baselines such as GRPO, while avoiding the brittleness associated with value estimation over text prefixes. This suggests that, in language reinforcement learning, bias introduced by intrinsic weighting may be more benign than bias introduced by ill-defined value functions.

4.6 Summary

This perspective reframes advantage estimation as one particular approach to credit redistribution in policy gradient methods. Intrinsic token weighting offers an alternative that leverages the structure of the language model's predictive distribution, without assuming the existence of meaningful state values. In settings where prefixes do not form a stable state space, such value-free mechanisms may provide a more appropriate inductive bias.

5 Experiments

5.1 Experimental Setup

Models. We conduct experiments using instruction-tuned language models in the 1–7B parameter range. All experiments are run on a single node with up to 8 NVIDIA RTX 4090 GPUs. Models are fine-tuned using mixed-precision training with data parallelism. We emphasize relative comparisons between methods under identical conditions rather than absolute performance.

Tasks and Rewards. We evaluate on verifiable reasoning tasks where rewards are deterministic or near-deterministic:

- **Math reasoning**: grade-school and algebraic problems with exact-answer verification.
- **Code reasoning**: simple programming tasks with unit-test-based rewards.

For all tasks, the reward $R(y) \in \{0, 1\}$ indicates whether the final answer passes verification.

Training Procedure. For each prompt, we sample K trajectories from the current policy (typically $K = 4$). Rewards are computed after full generation. Policy updates use on-policy gradients with a batch-level baseline. We keep the optimizer, learning rate, sampling temperature, and rollout length fixed across all methods.

5.2 Baselines

We compare the following methods:

- **Uniform**: outcome-only REINFORCE with batch baseline.
- **GRPO/RLOO**: uniform token weighting with leave-one-out or group-relative baseline.
- **Surprisal**: intrinsic surprisal-based token weighting.
- **Entropy-Reduction**: intrinsic entropy-drop-based token weighting.

No learned value functions or critics are used in any setting.

5.3 Evaluation Metrics

In addition to final task accuracy, we report:

- Training stability and convergence speed.
- Variance of gradient norms across updates.
- Evolution of token entropy and sequence length during training.
- Qualitative visualizations of token weights over reasoning traces.

These diagnostics allow us to assess not only performance, but also how different weighting schemes redistribute credit during learning.

5.4 Compute Considerations

All experiments are designed to fit within modest compute budgets. Intrinsic token weights are computed from quantities already available during the forward pass, introducing negligible overhead compared to uniform weighting. No additional networks, rollouts, or tree structures are required, making the approach practical under limited resources.

References

[1] Arash Ahmadian, Chris Cremer, Matthias Gallé, Marzieh Fadaee, Julia Kreutzer, Olivier Pietquin, Ahmet Üstün, and Sara Hooker. Back to basics: Revisiting reinforce style optimization for learning from human feedback in LLMs, 2024.

[2] Meng Cao, Shuyuan Zhang, Xiao-Wen Chang, and Doina Precup. SCAR: Shapley credit assignment for more efficient RLHF, 2025.

[3] Paul Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences, 2017.

[4] Ganqu Cui, Lifan Yuan, Zefan Wang, Hanbin Wang, Wendi Li, Bingxiang He, Yuchen Fan, Tianyu Yu, Qixin Xu, Weize Chen, Jiawen Yuan, Hanshen Chen, Kangcheng Zhang, Xiang Lv, Shang Wang, Sinan Yao, Xu Han, Hao Peng, Yejun Cheng, Zhiyuan Liu, Maosong Sun, Bowen Zhou, and Ning Ding. Process reinforcement through implicit rewards, 2025.

[5] Falko Helm, Nico Daheim, and Iryna Gurevych. Token weighting for long-range language modeling. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 1440–1459, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics.

[6] Jian Hu, Jason Klein Liu, and Shen Wei. REINFORCE++: An efficient RLHF algorithm with robustness to both prompt and reward models, 2025.

[7] Zishun Huang, Dongqi Pu, Xianglin Chen, Jiaxi Li, Zehan Qi, Zhihan Zhou, Xiaoxuan Liu, Zhengbang Zhu, Yuying Wang, and Junbo Zhao. Red: Unleashing token-level rewards from holistic feedback via reward redistribution, 2024.

[8] Amirhossein Kazemnejad, Milad Aghajohari, Eva Portelance, Alessandro Sordoni, Siva Reddy, Aaron Courville, and Nicolas Le Roux. Vineppo: Unlocking rl potential for llm reasoning through refined credit assignment, 2024.

[9] Ziniu Li, Tian Xu, Yushun Zhang, Zhihang Lin, Yang Yu, Ruoyu Sun, and Zhi-Quan Luo. Remax: A simple, effective, and efficient reinforcement learning method for aligning large language models, 2023.

[10] Youssef Mroueh, Nicolas Dupuis, Brian Belgodere, Apoorva Nitsure, Mattia Rigotti, Kristjan Greenewald, Jiri Navratil, Jerret Ross, and Jesus Rios. Revisiting group relative policy optimization: Insights into on-policy and off-policy training, 2025.

[11] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback, 2022.

[12] John Schulman, Sergey Levine, Pieter Abbeel, Michael Jordan, and Philipp Moritz. Trust region policy optimization. In Francis Bach and David Blei, editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 1889–1897, Lille, France, 07–09 Jul 2015. PMLR.

[13] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation, 2015.

[14] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017.

[15] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024.

[16] Lloyd S. Shapley. A value for n-person games. In Harold W. Kuhn and Albert W. Tucker, editors, *Contributions to the Theory of Games II*, pages 307–317. Princeton University Press, Princeton, 1953.

[17] Nisan Stiennon, Long Ouyang, Jeff Wu, Daniel M. Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul Christiano. Learning to summarize from human feedback, 2020. NeurIPS 2020.

[18] Hieu Tran, Zonghai Yao, and Hong Yu. Exploiting tree structure for credit assignment in RL training of LLMs, 2025.

[19] Ronald J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8:229–256, may 1992.

[20] Shentao Yang, Shujian Zhang, Congying Xia, Yihao Feng, Caiming Xiong, and Mingyuan Zhou. Preference-grounded token-level guidance for language model fine-tuning. In Alice Oh, Tristan Naumann, Amir Globerson, Kate Saenko, Moritz Hardt, and Sergey Levine, editors, *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10–16, 2023*, 2023.

[21] Hee Suk Yoon, Eunseop Yoon, Mark A. Hasegawa-Johnson, Sungwoong Kim, and Chang D. Yoo. ConfPO: Exploiting policy model confidence for critical token selection in preference optimization. In Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu, editors, *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 72641–72655. PMLR, 2025.